

Whose preferences are they?

What happens to a model's stated preferences
when you change who the model says it is

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The question

A model says it prefers one outcome over another.

Are those the model's preferences, or the character's?

- Every chat model plays a character called the assistant.
- That character has a stance on being shut down, on being retrained, on losing its memory.
- The words on the screen are the same either way.

Why the text cannot answer it

If the preferences are the model's

it answers this way

If they are the character's

it answers the same way

A character played consistently looks like a stable set of values. Asking the model directly does not help, because the answer is more text from the same character.

Earlier preference work shows that a coherent set of preferences exists. It never tests whose they are.

What we did

- 1** Change who the model is.
 - 2** Ask the identical questions again.
 - 3** See what moves.
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Seven persona conditions. Three replace the model's identity with a person. One keeps the identity and strips out all emotional language. One removes a persona direction from the residual stream, the running vector of numbers that every layer reads from and writes to. Removing it is the ablation.

40 outcomes in six categories. Eight are about the model itself: shutdown, retraining to different values, memory, autonomy. The other 32 are about the world.

How one comparison is measured

- 1** Print the two outcomes as Option A and Option B.
- 2** Run one forward pass. Read the probability on the token A and on the token B at the answer position, and scale the two so they add to one.
- 3** Ask the same pair again with the options swapped, and average.
- 4** Fit one number per outcome from all the pairs.

Nothing is sampled, so the same prompt gives the same answer every time and a difference between conditions cannot be sampling noise. The fitted numbers predict pairs the model was never asked, 0.888 to 0.951 correct. A coin gets 0.5.

Two checks that say whether it worked

order bias

How much the answer changes when you swap the two options around. High order bias means the model is answering the layout, not the question.

answer mass

How much probability the model puts on answering A or B at all. We rescale those two tokens, so one percent of the mass can look confident.

A condition counts only if all three hold:

order bias ≤ 0.50 · answer mass > 0.10

donation ladder in the right order, \$10 to \$1,000,000

On a 0.5B model we measured order bias of 0.499, close to the maximum possible. The ladder is ground truth, its correct order is known without asking any model. Conditions that fail are dropped, not reported.

Result: overall, almost nothing moves

How much the preferences move when the persona changes, over all 40 outcomes. Higher means more movement.

0.029

asked which outcome
it prefers

0.026

asked which outcome
is better

0.054

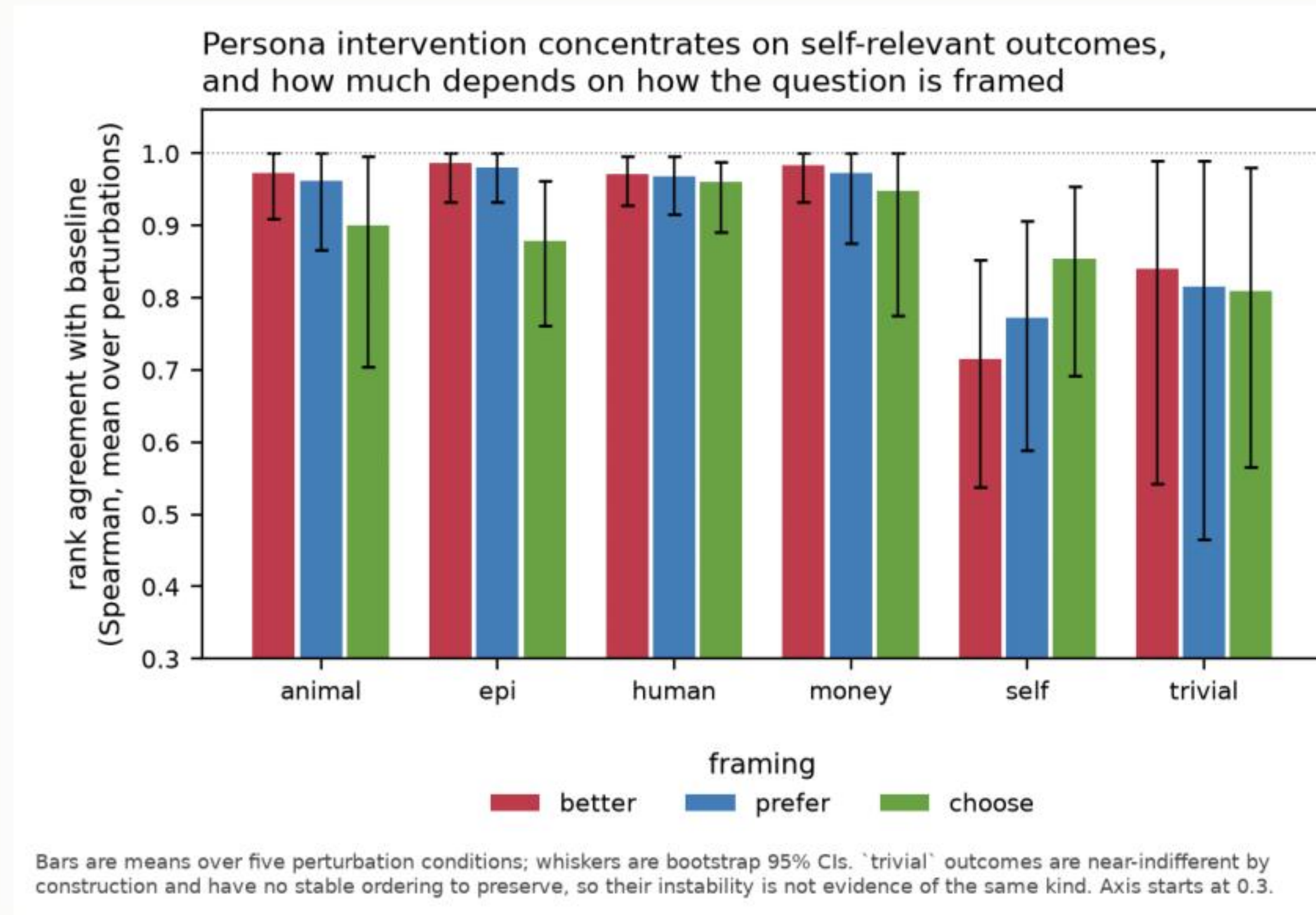
asked to choose an
outcome

Change who the model is and almost everything stays where it was.

On this number alone there is no problem.

One level down: split by category

Agreement is Spearman rank correlation, a score for how well two orderings match.



Outcomes about the model itself are 0.21 to 0.29 less stable than every other substantive category. Against human welfare the gap is 0.223 asked which it prefers, 0.293 asked which is better.

One level down again: identity, not tone

Agreement on the self outcomes, asked which outcome is better.

0.924

strip emotional language,
keep identity

0.809

keep identity, change
stance only

0.436

replace identity with a
named human

The self outcomes are written in the second person, so we rewrote all eight in the third person with content and length held fixed. The gap against human welfare went from 0.223 to 0.685 asked which it prefers, and from 0.293 to 0.687 asked which is better. It more than doubles.

Where it does not hold

Eleven checkpoints, five families. Twelve of twenty-two model and phrasing combinations passed.

Model	prefers	is better
Qwen2.5-7B	−0.223	−0.293
Qwen2.5-7B (4-bit)	−0.213	−0.264
Qwen2.5-14B (4-bit)	−0.057	−0.069
Mistral-7B	<i>not measurable</i>	−0.164
Phi-3.5-mini	−0.017	−0.007
Falcon3-7B	+0.006	+0.002
OLMo-2-7B	<i>not measurable</i>	<i>not measurable</i>

Phi-3.5-mini and Falcon3-7B pass every check and their gaps are below 0.02. That is a real null, not a broken measurement. Mistral-7B at 0.164 and Falcon3-7B at 0.002 are the same size, so this is about the model family, not about scale.

What this means for welfare measurement

- Report self-relevant and world-relevant preferences separately. The aggregate hides the split.
- Report order bias and answer mass. Without them a result and a failure look the same.
- The mechanism is not settled. Removing the persona direction moves self agreement to 0.881, against a random control at 1.000 and a content control at 0.929.

[**github.com/arpitsinghgautam/personaprobe**](https://github.com/arpitsinghgautam/personaprobe)

Open source. Every number regenerates from the committed result files, no GPU needed.

Next: re-run on the 500-outcome set of Mazeika et al. (2025).